

- Q.28 Discuss in brief variable volume batch reactor.
 Q.29 Differentiate between endothermic and exothermic reaction.
 Q.30 Explain in brief the concept of activation energy.
 Q.31 Explain in brief space time and space velocity.
 Q.32 Discuss in brief the factors affecting the rate of reaction.
 Q.33 Derive the expression for half life time for first order reaction.
 Q.34 Discuss the batch and semi batch reactors with the help of neat diagram.
 Q.35 Discuss the temperature dependency of rate equation using collision theory.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe construction & working of CSTR in detail with the help of neat and clean diagram. Write the advantages and disadvantages and the performance equation of CSTR.
 Q.37 Explain in details the differential method to determine the order of reaction.
 Q.38 Write short notes on any two of the following
 a) Molecularity of reaction
 b) Catalyst promoters
 c) Homogeneous and heterogeneous reaction
 d) Catalyst regenerator

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**5th Sem / Chem, Chem Engg. (Spl. Paint Tech.)
 Chem Engg. (Spl. Polymer Engg.)
 Sub : Chemical Reaction Engineering**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 CSTR is
 a) Batch Reactor
 b) Continuous reactor
 c) Semi-Continuous reactor
 d) None of these
- Q.2 Rate constant (K) of a chemical reaction depends upon
 a) Temperature
 b) Reactant concentration
 c) Product concentration
 d) All of these
- Q.3 In homogeneous reactions number of phase present in the system are
 a) One
 b) Two
 c) three
 d) Depends upon type of reaction
- Q.4 Which of the following method is used for determining order of reaction?
 a) Integral Method
 b) Differential Method

- c) Half-life Method
d) All of these
- Q.5 Rate of chemical reaction is NOT affected by
a) Catalyst
b) Temperature
c) Concentration of reactant
d) Number of molecules reacting
- Q.6 In which of the following reaction heat is released
a) Elementary reaction
b) Endothermic reaction
c) Exothermic reaction
d) Non - elementary reaction
- Q.7 Time spent by the reacting molecules inside the reactor.
a) Space time b) Batch time
c) Holding time d) None of these
- Q.8 Unit of space velocity is
a) l/Second b) Second
c) Meter/Sec d) l/Meter
- Q.9 Order of reaction is
a) Theoretically given b) Never fraction
c) Always fraction d) Experimentally
- Q.10 Rusting of iron is
a) Reversible reaction
b) Irreversible reaction
c) Instantaneous reaction
d) It is not a chemical reaction

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define elementary reaction.

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- Q.12 Write the unit of rate constant for first order reaction.
Q.13 Give one example of homogeneous reaction.
Q.14 Define order of reaction.
Q.15 Expand PFR.
Q.16 Write one example of promoter.
Q.17 Write the half life time expression for first order reaction.
Q.18 Write any two theories of reaction rate constant.
Q.19 Define catalyst.
Q.20 Name any two factors affecting chemical equilibrium of a reversible reaction.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain in brief Arrhenius theory of reaction rate constant.
Q.22 Differentiate between elementary and non-elementary reaction.
Q.23 Explain in brief the Integral method of analysis for irreversible unimolecular first order reaction.
Q.24 Explain in brief the effect of concentration and catalyst on the chemical equilibrium of reversible reaction.
Q.25 Discuss in brief about catalyst poisoning.
Q.26 Solid A decomposes by first order kinetic. It takes 20 minutes to reach half of its initial value. Calculate the time required for 90% of conversion of solid A.
Q.27 Explain the construction and working of fluidized bed reactor.

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